What does a richer colour vocabulary buy us?

An analysis of the World Color Survey

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Same chips. Different names.
What can regression explain?

From counting colour words to understanding how people organise colour space

01 Vocabulary & geometry

Does a richer colour vocabulary predict finer distinctions between nearby colours and more compact colour categories, beyond simply using more labels?

02 The individual at the extremes

Are richer-vocabulary languages associated with fewer low-scoring speakers and more high-scoring speakers?

03 Age, sex & interactions

Do age and sex relate to colour categorisation? Does vocabulary richness have different associations across age groups and sexes, or across languages?

04 Where do differences occur?

Are richness and demographic associations stronger in particular regions of colour space?

05 Does the model learn anything useful?

Do vocabulary and demographic variables improve prediction of individual partition scores enough to justify model complexity?

From theory to broader evidence



Berlin & Kay

Basic Color Terms (1969) **Question:** Are colour categories entirely language-specific?
Their study of 20 languages proposed 11 basic colour categories and a constrained sequence in which languages tend to acquire additional basic terms.

Contribution: A theory of cross-linguistic regularities in colour naming, the size and representativeness of the original sample motivated wider testing.

World Color Survey

A larger empirical test The WCS extended the investigation to 110 languages and 2,616 informants, using the same standard array of 330 Munsell colour chips.
It recorded individual naming responses and focal-colour selections, allowing comparisons across languages and between speakers of the same language.

Contribution: A much broader dataset for testing recurring patterns while retaining individual variation.

From recurring motifs to the present analysis



Lindsey & Brown

PNAS (2009) Analysing 2,367 WCS informants, they identified recurring colour-naming motifs while finding that several different motifs often occurred within one language.
Their work shifted attention from one label system per language to the diversity of individual speakers' naming patterns.

Implication: Cross-linguistic regularities can coexist with substantial within-language variation.

The present project

Regression Techniques We ask whether vocabulary richness is associated with finer distinctions between nearby chips, more compact categories, and different proportions of low- and high-scoring speakers.
We then examine age, sex, colour region and whether added predictors improve held-out prediction enough to justify their complexity.

Contribution: Quantify associations using course-level regression, diagnostics and model comparison.

What is in the World Color Survey?

110 languages

2,616 informants

330 Munsell chips per speaker

863,280 naming responses in archive

Two main tasks

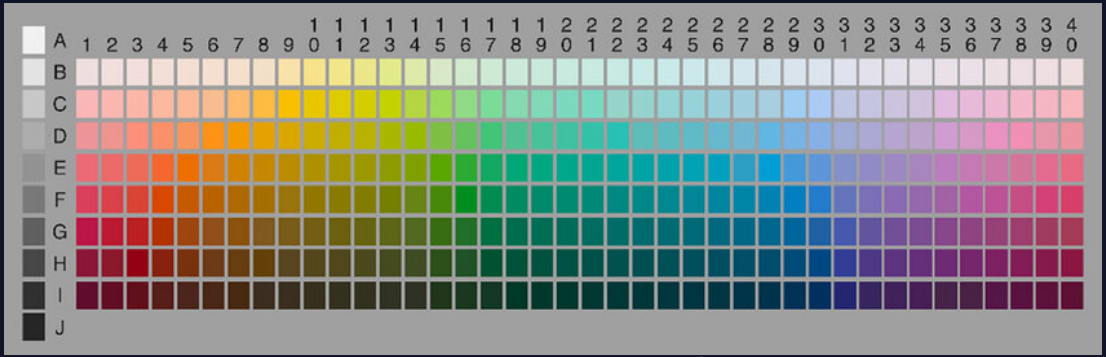
Naming task: For each chip, record the colour term supplied by the speaker.

Focal-colour task: For each major term, select its best example colour chip or chips.

Which speakers enter the analysis?

- **2,447 speakers:** We kept speakers who named at least 300 of the 330 colour chips.
- **109 languages:** After this filter, at least one eligible speaker remained in 109 of the original 110 languages.
- **2,355 speakers:** For the main regression involving age and sex, we further retained only speakers with non-missing outcome, age and sex data.

These are the colour chips used in the study



These are the 330 Munsell colour chips used in the World Color Survey. In the naming task, speakers supplied a colour term for each chip. In the separate focal-colour task, they selected the best example chip or chips for a colour term.

From archival files to analysis variables

Files used	Records × columns
`term.txt` Language, speaker, chip and assigned term code.	863,280 × 4
`chip.txt` Chip ID and its WCS chart coordinates.	330 × 4
`dict.txt` Maps language-specific term codes to recorded transcriptions.	2,364 × 4
`foci-exp.txt` Focal-colour selections, expanded to one row per selected chip.	110,784 × 5
`spkr-lsas.txt` Language, speaker ID, approximate age and sex, where available.	2,619 × 4
`cnum-vhcm-lab-new.txt` Chip-to-Munsell mapping and CIELAB coordinates (L^*, a^*, b^*).	330 × 9

Do richer colour vocabularies correspond to finer distinctions and more compact colour categories?



What exactly are we asking about vocabulary richness?

1. **Fine-grained lexical resolution**
   Do languages with a larger shared inventory of colour terms give different names to a larger fraction of nearby sampled colour-chip pairs?

2. **Beyond the number of labels**
   Does that association remain after accounting for how many distinct terms the individual speaker uses?

3. **Geometry of categories**
   Are richer inventories associated with more compact naming categories, after adjusting for the number of labels?

4. **Agreement between tasks**
   Are selected focal chips closer to the geometric centres of the categories that the same speakers named?

First test: what counts as shared vocabulary richness?

Question. Do languages with richer colour vocabularies partition colour space more finely?

Let ℓ index a language, s a speaker, and t a colour term. Let $\mathcal{S}_\ell$ be the set of eligible speakers in language ℓ, and let $n_\ell = |\mathcal{S}_\ell|$ be their number.

Count recurrent terms across speakers

For each language and term pair, define

$$N_{\ell t} = \#\{s \in \mathcal{S}_\ell : \text{speaker } s \text{ uses } t \text{ on at least one observed chip}\}.$$

The recurrent inventory size is

$$R_\ell = \sum_t \mathbf{1}\{N_{\ell t} \geq 2\}.$$

Here $\mathbf{1}\{A\} = 1$ if statement A is true and 0 otherwise. Thus, a term contributes to R_ℓ only when at least two eligible speakers use it.

Second richness measure: an individual's term count

Let $\mathcal{O}_s$ be the set of colour chips for which speaker s supplied a naming response. For $i \in \mathcal{O}_s$, let Y_{si} be the term assigned by speaker s to chip i.

Individual vocabulary size

$$K_s = \left|\{Y_{si} : i \in \mathcal{O}_s\}\right|.$$

K_s counts distinct terms used by speaker s on observed chips.

Peer estimate for a held-out speaker

For target speaker s in language ℓ, the mean term count among the other eligible speakers is

$$\bar{K}_{\ell,-s} = \frac{1}{n_\ell - 1} \sum_{\substack{r \in \mathcal{S}_\ell \\ r \neq s}} K_r.$$

Excluding s prevents their own responses from contributing to this predictor.

R_ℓ counts recurrent terms. K_s counts terms used by one speaker. $\bar{K}_{\ell,-s}$ averages individual counts over peers.

Shared richness for a held-out speaker

For a target speaker s in language ℓ, define the peer set

$$\mathcal{S}_{\ell,-s} = \mathcal{S}_\ell \setminus \{s\}.$$

For each term t, count how many of those other speakers use it on at least one observed chip:

$$N_{\ell t,-s} = \#\{r \in \mathcal{S}_{\ell,-s} : \exists i \in \mathcal{O}_r \text{ with } Y_{ri} = t\}.$$

The leave-one-speaker-out recurrent inventory is

$$R_{\ell,-s} = \sum_t \mathbf{1}\{N_{\ell t,-s} \geq 2\}.$$

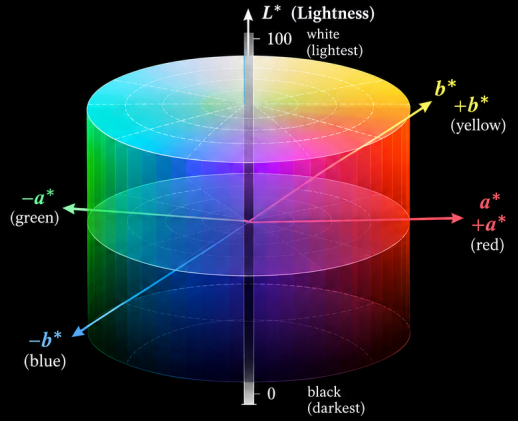
Why exclude the target speaker?

When $R_{\ell,-s}$ predicts speaker s's outcome, that speaker's own naming responses do not help construct the predictor. A term counts only if at least two of the *other* eligible speakers use it.

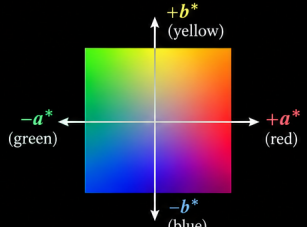
$R_{\ell,-s}$ counts the number of recurrent terms among peers and $\bar{K}_{\ell,-s}$ counts the average number of distinct terms peers use.

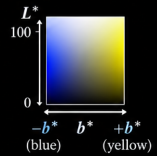
The CIELAB colour space (L^*, a^*, b^*)

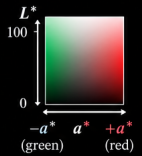
A 3D, approximately perceptually–uniform colour space











Example colours (L^*, a^*, b^*)

$L^* = 70$ $a^* = 30$ $b^* = 10$	$L^* = 60$ $a^* = -20$ $b^* = -40$	$L^* = 80$ $a^* = -10$ $b^* = 60$	$L^* = 30$ $a^* = 20$ $b^* = -30$

Distance in CIELAB

Euclidean distance in (L^*, a^*, b^*) correlates reasonably well with perceived colour difference:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$$

Key points

- L^*: lightness (0 = black, 100 = white)
- a^*: green (−) ↔ red (+)
- b^*: blue (−) ↔ yellow (+)
- Approximately perceptually uniform (equal distances ≈ equal perceived differences, though not perfect)

Metric 1a: build the local neighbour graph

Each chip i has CIELAB coordinates $\mathbf{x}_i = (L_i^*, a_i^*, b_i^*)$. For chips i and j, their CIE 1976 Euclidean distance is

$$\Delta E_{76}(i,j) = \sqrt{(L_i^* - L_j^*)^2 + (a_i^* - a_j^*)^2 + (b_i^* - b_j^*)^2}.$$

From distances to unique pairs

Let $\mathcal{N}_3(i)$ be the set of the three chips closest to chip i under this distance. Build

$$\mathcal{E} = \big\{\{i,j\} : j \in \mathcal{N}_3(i) \text{ or } i \in \mathcal{N}_3(j)\big\}.$$

Each element is an **unordered pair**: $\{i,j\} = \{j,i\}$, so the same pair is retained only once even if both chips select each other. The relationship need not be mutual for the pair to be included.

$k = 3$ is a design choice to represent local colour neighbours.

Metric 1b: turn neighbouring pairs into a speaker score

Let $\mathcal{O}_s$ be the set of chips named by speaker s, and let Y_{si} be the term assigned to observed chip i. Retain only neighbour pairs for which both names were observed:

$$\mathcal{E}_s = \{\{i, j\} \in \mathcal{E} : i, j \in \mathcal{O}_s\}.$$

For each observed pair, define

$$D_{sij} = \mathbf{1}\{Y_{si} \neq Y_{sj}\}.$$

Thus $D_{sij} = 1$ if the labels differ and 0 if they agree. The local lexical-resolution score is

$$Q_s^{\text{split}} = \frac{\sum_{\{i,j\} \in \mathcal{E}_s} D_{sij}}{|\mathcal{E}_s|}.$$

Interpretation

Q_s^{split} is the fraction of observed neighbouring pairs that speaker s names differently. A higher value means finer lexical differentiation on this fixed neighbour graph.

The score is undefined if $|\mathcal{E}_s| = 0$ and such a case will be omitted in this metric's analysis.

Metric 2a: naming categories and centroids

For speaker s, let $\mathcal{O}_s$ be the set of chips with an observed name. For each $i \in \mathcal{O}_s$, Y_{si} is the term assigned to chip i, and $\mathbf{x}_i = (L_i^*, a_i^*, b_i^*)$ is its CIELAB coordinate vector.

The chips assigned to a term

For a term t used by speaker s, define the category's chip set

$$I_{st} = \{i \in \mathcal{O}_s : Y_{si} = t\}.$$

Its size is $|I_{st}|$. If $|I_{st}| > 0$, the category's CIELAB centroid is

$$\boldsymbol{\mu}_{st} = \frac{1}{|I_{st}|} \sum_{i \in I_{st}} \mathbf{x}_i.$$

- I_{st} contains exactly the chips speaker s names with term t.
- $\boldsymbol{\mu}_{st}$ is their coordinate-wise mean. It does not need to coincide with a displayed chip.
- An unused term has no centroid for that speaker and is not included in the calculation.

Metric 2b: quantify within-category dispersion

Use $\mathcal{O}_s$, Y_{si}, $\mathbf{x}_i$ and the category centroids $\boldsymbol{\mu}_{st}$ defined on the preceding frame. Let $m_s = |\mathcal{O}_s|$ be the number of chips named by speaker s.

Root-mean-square within-term distance

$$W_s = \sqrt{\frac{1}{m_s} \sum_{i \in \mathcal{O}_s} \left\| \mathbf{x}_i - \boldsymbol{\mu}_{s, Y_{si}} \right\|_2^2}.$$

Here $\| \cdot \|_2$ is Euclidean length in the three CIELAB dimensions. Smaller W_s means the named chips lie closer to their assigned category centroids.

Set $U_s = -W_s$. Let $\bar{U}$ and s_U be the sample mean and standard deviation of U_s across eligible speakers with defined scores. Then

$$Z_s^{\text{compact}} = \frac{U_s - \bar{U}}{s_U}.$$

P.S. U_s has been set to $-W_s$ so that higher values of U_s correspond to greater compactness. Higher Z_s^{compact} means smaller within-term dispersion. If every observed chip has its own term, $W_s = 0$.

Metric 2c: alignment with focal colours

For speaker s, term t, and chip i, use the category set I_{st} and centroid $\boldsymbol{\mu}_{st}$ defined earlier. Let $\mathcal{F}_s$ be the set of usable pairs (t, i) in which speaker s selected chip i as a focal example of term t and $|I_{st}| > 0$.

Mean focal-to-centroid distance

For $|\mathcal{F}_s| > 0$, define

$$D_s^{\text{focal}} = \frac{1}{|\mathcal{F}_s|} \sum_{(t,i)\in\mathcal{F}_s} \|\mathbf{x}_i - \boldsymbol{\mu}_{st}\|_2.$$

Let $V_s = -D_s^{\text{focal}}$, and let $\bar{V}, s_V$ be its mean and sample standard deviation over eligible speakers with defined scores. Then

$$Z_s^{\text{focal}} = \frac{V_s - \bar{V}}{s_V}.$$

A high score means a speaker's selected focal chips lie closer to the centroid of chips they name with the corresponding term. This measures alignment between two WCS tasks.

Assumptions behind the retained OLS analyses

For the OLS analyses retained in the project, the response $\bar{Q}_\ell$ is the mean of a selected speaker-level metric Q_s among eligible speakers in language ℓ. The model uses one row per language, so the regression error is language-level ε_ℓ.

Working linear model

$$\bar{Q}_\ell = \mathbf{z}_\ell^\top \beta + \varepsilon_\ell,$$

where $\mathbf{z}_\ell$ is the vector of predictors (including an intercept), β is the corresponding coefficient vector, and ε_ℓ is the error for language ℓ.

- **Mean specification:** the formula should describe the conditional mean adequately.
- **Independence:** language-row errors are treated as independent, though historical or geographic factors may still link languages.
- **Constant variance and normality:** these support conventional small-sample standard errors and tests. Residual diagnostics assess plausibility, not truth.
- **Comparable samples:** each model comparison must use the same response and language rows.

The primary language-mean vocabulary sequence below instead uses ridge and lasso. OLS remains in the tail, demographic and regional analyses.

How fit and complexity are compared

Let n be the number of observations, k the number of parameters counted by the likelihood criterion, and $\widehat{\mathcal{L}}$ the maximised likelihood. The information criteria are

$$AIC = -2\log\widehat{\mathcal{L}} + 2k, \qquad BIC = -2\log\widehat{\mathcal{L}} + k\log n.$$

Lower AIC/BIC is preferred only for models fitted to the same response and observations. BIC generally penalises extra parameters more strongly when n is large.

For observed responses y_i, let $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$, and let $\hat{y}_i$ be the fitted value. Then

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

For p predictors excluding the intercept,

$$R^2_{\mathrm{adj}} = 1 - \frac{n-1}{n-p-1}(1 - R^2).$$

PRESS: leave-one-out prediction

Let n be the number of observations, y_i the observed response for row i, and $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$ their sample mean. Let $\hat{y}_{i,(-i)}$ be the prediction for row i from the same OLS formula refitted after omitting that row. The leave-one-out error is $e_{i,(-i)} = y_i - \hat{y}_{i,(-i)}$.

$$PRESS = \sum_{i=1}^{n} e_{i,(-i)}^2 = \sum_{i=1}^{n} (y_i - \hat{y}_{i,(-i)})^2.$$

Two scale-aware summaries are

$$R^2_{PRESS} = 1 - \frac{PRESS}{\sum_{i=1}^{n}(y_i - \bar{y})^2}, \qquad RMSE_{LOO} = \sqrt{\frac{PRESS}{n}}.$$

For a model comparison, let $C_{\text{candidate}}$ and C_{simpler} be values of the same criterion for the candidate and simpler model, fitted to the same response and rows. Define

$$\Delta C = C_{\text{candidate}} - C_{\text{simpler}}.$$

Negative ΔAIC, ΔBIC, or $\Delta PRESS$ favours the candidate on that criterion.

Lower PRESS/RMSE$_{LOO}$ is better and negative R^2_{PRESS} means held-out predictions are worse than the mean benchmark. The ridge/lasso language-mean models below use repeated nested cross-validation instead.

Language means: outcome and predictors

Unit of analysis. One row represents one language ℓ. The two primary responses are fitted separately.

$$\bar{Q}_\ell^{\text{split}} = \text{mean near-split score in language } \ell, \qquad \bar{Q}_\ell^{\text{compact}} = \text{mean standardised compactness score.}$$

Predictors

- $\log(n_\ell)$ adjusts for the number of eligible speakers.
- R_ℓ counts recurrent colour terms used by at least two eligible speakers.
- $\bar{K}_\ell$ is the mean number of distinct colour terms used by an eligible speaker.

The response is a language mean. The predictor $\bar{K}_\ell$ is derived from the same naming task and is therefore a concurrent explanatory variable rather than a pre-response predictor.

Ridge and lasso: predictor sequence for language means

Each outcome is fitted separately using one row per language. The three predictor sets are

$$
\begin{aligned}
M_0: &\quad \log(n_\ell), \\
M_1: &\quad \log(n_\ell) + R_\ell, \\
M_2: &\quad \log(n_\ell) + R_\ell + \bar{K}_\ell.
\end{aligned}
$$

Here n_ℓ is the eligible-speaker count, R_ℓ is recurrent inventory richness, and $\bar{K}_\ell$ is mean individual term count.

The two penalties

Ridge

$$\mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

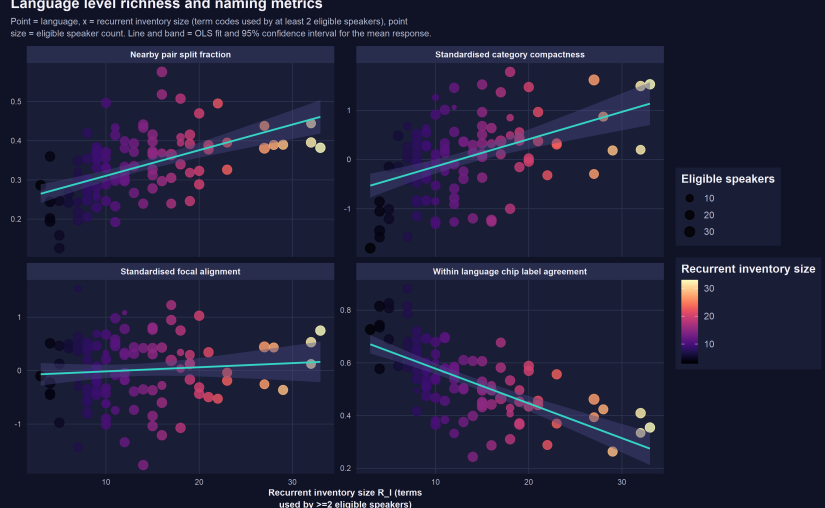
Shrinks correlated coefficients towards zero.

Lasso

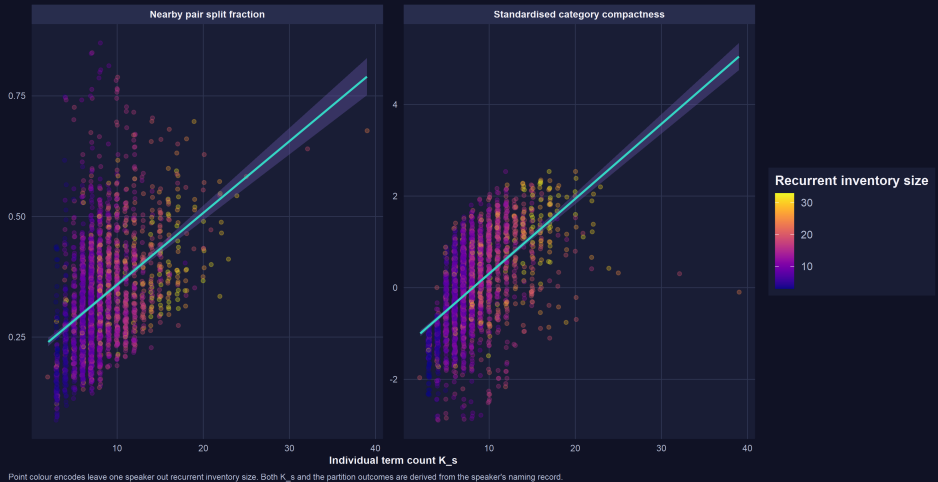
$$\mathrm{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|$$

Can set some coefficients exactly to zero.

Lambda is selected by five-fold inner cross-validation inside each training set. Performance is assessed over five repeats of outer five-fold cross-validation, with all 109 languages predicted in each repeat. The supplied CV summary contains M1 and M2 scores only. The one-predictor M_0 ridge/lasso fits returned errors, so no M_0 RMSE is available and no M_1-versus-M_0 CV conclusion is made.



Each dot is a language, sized by eligible-speaker count. The horizontal axis and colour show recurrent inventory size. Lines and bands are OLS trends with 95% confidence intervals. Panels show near-split, standardised compactness, focal-to-centroid alignment, and the language mean of per-chip agreement with other speakers. Focal alignment and agreement are descriptive context; formal models focus on the first two outcomes.



Each dot is a language. This sensitivity plot relates mean individual term count to the naming metrics. The association is especially strong for compactness, so term count is interpreted as a concurrent summary of the same naming task rather than a predictor available before the task.

Results: what improves held-out prediction?

Outcome	Penalty	M1 RMSE (SD)	M2 RMSE (SD)	ΔRMSE	R^2_{pred} M1 → M2
Near-split	Lasso	0.06969 (0.00142)	0.07040 (0.00126)	+0.00071	0.238 → 0.222
Near-split	Ridge	0.06975 (0.00147)	0.07019 (0.00133)	+0.00045	0.237 → 0.227
Compactness	Lasso	0.70256 (0.00769)	0.53523 (0.00879)	−0.16734	0.187 → 0.528
Compactness	Ridge	0.70069 (0.00677)	0.54177 (0.00776)	−0.15892	0.192 → 0.517

- For near-split, adding mean individual term count in M2 slightly increases RMSE under both penalties. It does not improve held-out prediction over M1.
- For compactness, M2 reduces RMSE by about 0.159 to 0.167 and raises predictive R^2 from about 0.19 to 0.52 to 0.53.

Each repeat predicts all 109 languages. Lower RMSE is better. Parentheses give the standard deviation across the five outer-CV repeats.

Reading the regularised vocabulary coefficients

The table gives mean coefficients from the M2 models, expressed per one standard deviation of each predictor across the outer training folds.

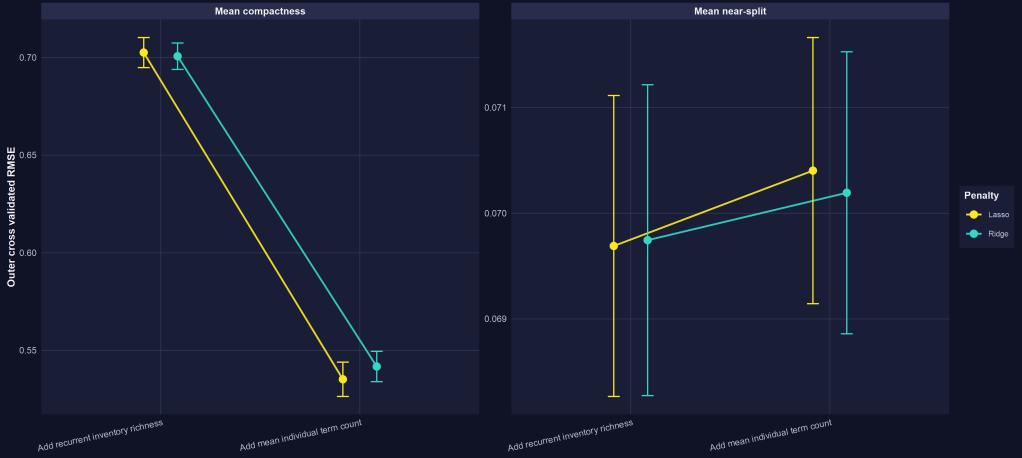
Outcome	Penalty	Recurrent richness R_ℓ	Mean terms $\bar{K}_\ell$
Near-split	Ridge	$+0.0238$	$+0.0151$
Near-split	Lasso	$+0.0296$	$+0.0105$
Compactness	Ridge	-0.1616	$+0.6447$
Compactness	Lasso	-0.2939	$+0.7953$

- For near-split, both vocabulary coefficients remain positive under both penalties.
- For compactness, mean individual term count has a large positive coefficient, while the richness coefficient turns negative after conditioning on mean term count.
- The two language-level vocabulary measures are strongly correlated ($r = 0.831$). The conditional sign reversal is therefore a shared-information issue, not evidence that richer inventories have a negative marginal association with compactness.

Lasso retains mean term count in 84% of near-split outer fits and 100% of compactness outer fits.

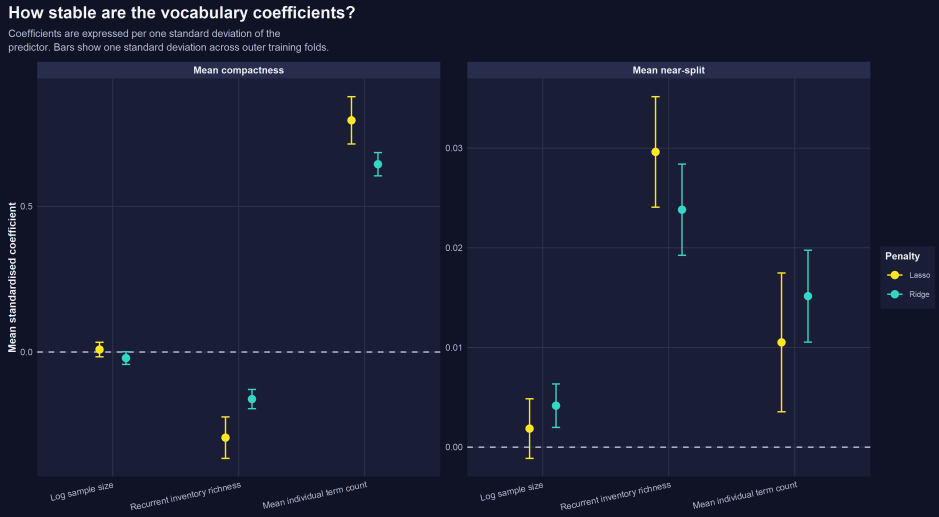
Ridge and lasso: language level prediction

Points show mean outer fold RMSE across repeated nested cross validation, bars show one standard deviation across repeats. Lower RMSE is better.



Each sequence starts with log eligible speaker count, adds recurrent inventory richness, then adds mean individual term count. Lambda is tuned within each outer training sample using inner cross validation. These outcomes and language summaries are derived from the naming task, so this is an explanatory within dataset prediction check rather than a pre naming forecast.

Points show mean outer-fold RMSE over five repeated nested five-fold cross-validation runs. Bars show one standard deviation across repeats. M1 contains log eligible-speaker count and recurrent richness. M2 additionally contains mean individual term count. Lower RMSE is better. M0 scores are absent from the supplied output, so this plot compares M1 with M2 only.

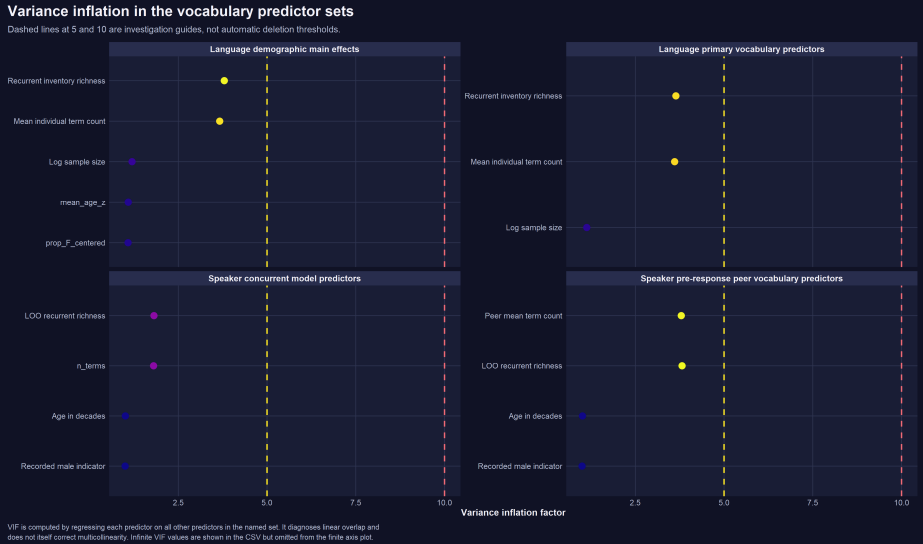


The plotted coefficients are from M2 and are expressed per one standard deviation of each predictor. Bars show one standard deviation across outer training folds, not confidence intervals. The negative compactness coefficient for recurrent richness is conditional on mean term count, which is strongly correlated with richness.

The vocabulary measures share information

Predictor pair	Pearson r	VIF (richness)	VIF (mean terms)
Language: R_ℓ and $\bar{K}_\ell$	0.831	3.64	3.61
Speaker peers: $R_{\ell,-s}$ and $\bar{K}_{\ell,-s}$	0.859	3.82	3.80

The simple vocabulary predictor sets have VIFs below the common value of 5 used as an investigation guide. However, the high pairwise correlations show that recurrent richness and mean term count carry overlapping information. These values do not describe the interaction-expanded models. Their separate diagnostics are higher: individual near-split VIF reaches 49.03. Interaction coefficients therefore need particular caution.



The plotted VIFs cover the simple language and speaker predictor sets. All displayed values are below 5. The interaction-expanded designs are not included in this figure and have higher VIFs, reaching 49.03 for the individual near-split interaction model.



Cells give pairwise Pearson correlations. Recurrent inventory richness and mean individual term count correlate 0.83 across languages. Their leave-one-speaker-out peer versions correlate 0.86. Pairwise correlation describes two predictors at a time, while VIF accounts for all other predictors in its named set.

From the first question to the next

What we learnt

- Adding mean term count does not improve near-split prediction over the sample-size-plus-richness M1 model.
- The same addition substantially improves compactness prediction.
- Recurrent richness and mean individual term count overlap, so compactness coefficients are conditional on their shared information.

The next question

An average might improve while a subset of speakers remains at the bottom, or richness might be related to both tails.

So we move from mean outcomes to the distribution: who appears in the low and high deciles?

Does richness change who falls in the tails?



Turning speaker scores into tail proportions

Choose a speaker-level metric Q_s (for example, Q_s^{split} or Z_s^{compact}). Let $q_{.10}$ and $q_{.90}$ be its empirical 10th and 90th percentiles among eligible speakers with a defined metric.

- **Speaker-level tail indicators**

  Define

  $$L_s = \mathbf{1}\{Q_s \le q_{.10}\}, \qquad H_s = \mathbf{1}\{Q_s \ge q_{.90}\}.$$

  $L_s = 1$ denotes a bottom-decile score, while $H_s = 1$ denotes a top-decile score.

- **Language-level prevalence**

  For language ℓ, let $\mathcal{S}_\ell$ be the speakers with a defined metric and $n_\ell = |\mathcal{S}_\ell|$. Define the numbers of lower-tail and upper-tail speakers as

  $$x_{\ell,L} = \sum_{s \in \mathcal{S}_\ell} L_s, \qquad x_{\ell,H} = \sum_{s \in \mathcal{S}_\ell} H_s.$$

  The corresponding language-level proportions are

  $$p_{\ell,L} = \frac{x_{\ell,L}}{n_\ell}, \qquad p_{\ell,H} = \frac{x_{\ell,H}}{n_\ell}.$$

Tail proportions: transformed linear model

For one selected tail, let x_ℓ be the number of speakers in that tail and n_ℓ the number eligible in language ℓ. The observed proportion is $\widehat{p}_\ell = x_\ell / n_\ell$.

Arcsine-square-root transformation

$$Z_\ell = \arcsin \sqrt{\widehat{p}_\ell}.$$

The language-level regression is

$$Z_\ell = \beta_0 + \beta_1 \log(n_\ell) + \beta_2 R_\ell + \varepsilon_\ell,$$

where R_ℓ is recurrent inventory richness, β_j are regression coefficients, and ε_ℓ is the language-level error.

Our response is a proportion of speakers in a particular tail. Proportions have a variance that depends on both the underlying proportion and the number of speakers observed. In particular, proportions around $\frac{1}{2}$ are more variable than proportions close to 0 or 1. We therefore apply the arcsine-square-root transformation. This transformation approximately cancels the dependence of the variance on the underlying proportion, making the response more suitable for linear regression.

Tail proportions: grouped-binomial GLM sensitivity

Let $X_\ell = x_\ell$ be the number of speakers in the chosen tail, n_ℓ the eligible denominator, and π_ℓ the modelled probability that an eligible speaker in language ℓ falls in that tail.

$$X_\ell \sim \text{Binomial}(n_\ell, \pi_\ell), \qquad \log \frac{\pi_\ell}{1-\pi_\ell} = \mathbf{z}_\ell^\top \beta.$$

Here $\mathbf{z}_\ell$ is the vector of predictors (including an intercept), and β is the coefficient vector. The logit link keeps fitted probabilities between 0 and 1.

- We model the observed integer success and failure counts directly, rather than approximating them with fractional prior weights.
- The binomial model assumes conditionally independent Bernoulli outcomes with a common probability within language.
- AIC/BIC comparisons use the same counts and analysis sample.

Baseline and added ladders: language-level tail prevalence

The analysis is fitted separately for near-split and compactness, and separately for the lower and upper tail. For language ℓ, $\bar{A}_\ell$ is mean recorded age in decades among speakers with age data, and F_ℓ is the proportion recorded female among speakers with sex data. These summaries are used in the grouped-binomial GLM sensitivity sequence.

A. Arcsine square root OLS

Response: $Z_\ell = \arcsin \sqrt{x_\ell / n_\ell}$.

1. TL_0: baseline with $\log(n_\ell)$.
2. TL_1: add recurrent inventory richness R_ℓ.
3. TL_2: add mean individual term count $\bar{K}_\ell$.

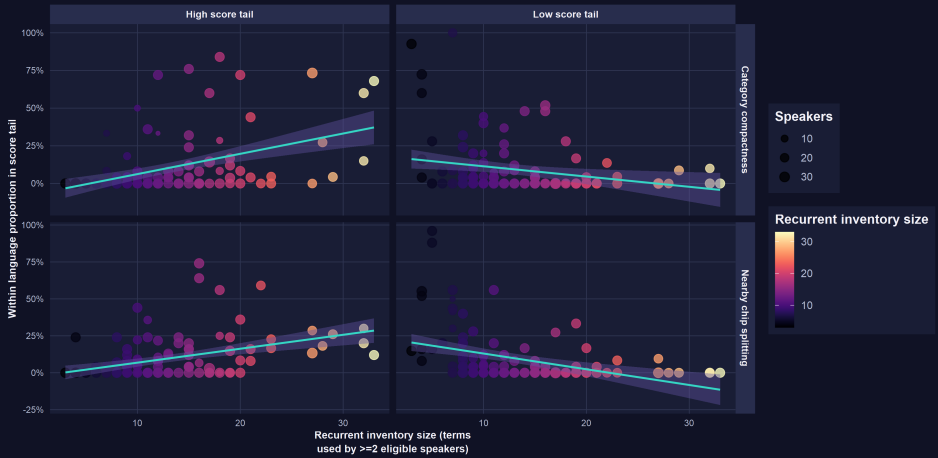
The result table reports the richness and mean term count steps.

B. Grouped binomial GLM sensitivity

Response: $X_\ell \sim \text{Binomial}(n_\ell, \pi_\ell)$ with a logit link.

1. G_0: intercept only.
2. G_1: add recurrent inventory richness R_ℓ.
3. G_2: add mean individual term count $\bar{K}_\ell$.
4. G_3: add $\bar{A}_\ell$ and F_ℓ.
5. G_4: add $\bar{A}_\ell R_\ell$, $F_\ell R_\ell$, and $\bar{A}_\ell F_\ell$.
6. G_5: add $\bar{A}_\ell F_\ell R_\ell$.

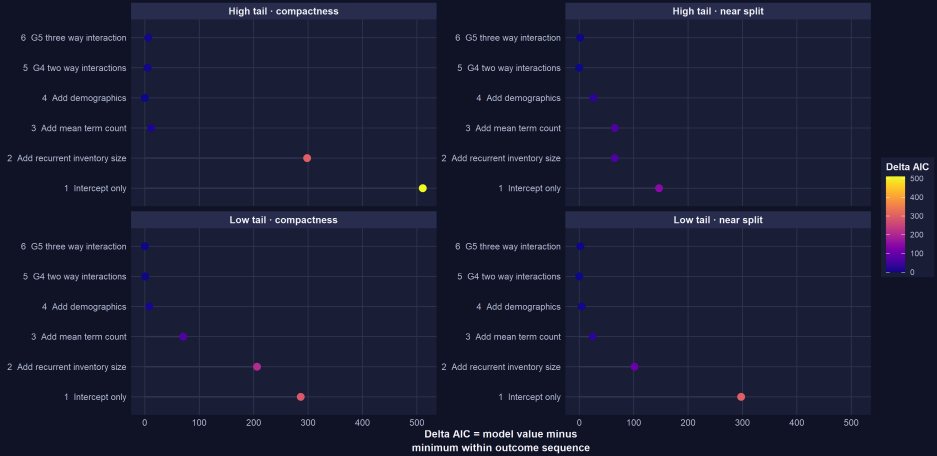
The AIC and BIC plots show the penalty comparison for this sequence.



Each dot is a language. The horizontal axis is recurrent inventory richness and the vertical axis is the observed proportion of speakers in the selected tail. Panels separate near split and compactness, then lower and upper tails. Read the direction separately for each outcome.

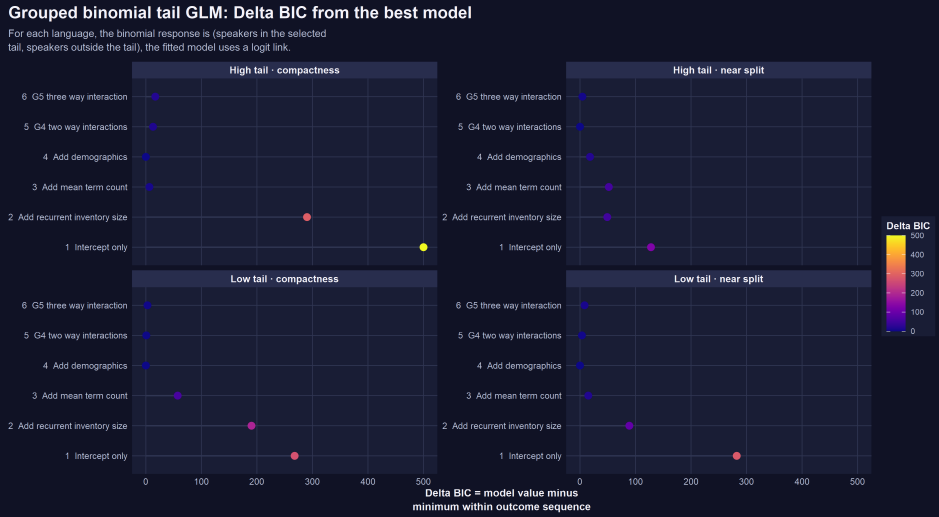
Grouped binomial tail GLM: Delta AIC from the best model

For each language, the binomial response is (speakers in the selected tail, speakers outside the tail), the fitted model uses a logit link.



Within each sequence, Delta AIC = AIC(model) minus minimum AIC. AIC = minus 2 log(L) + 2k, BIC = minus 2 log(L) + k log(n), where L is the maximised likelihood, k is the parameter count and n is the fitted sample size.

Each point is AIC minus the minimum within an outcome, so values near zero are preferred. G_4 has the lowest AIC for both near-split tails. G_5 is only slightly lower than G_4 for the compactness lower tail, while G_3 is lowest for the compactness upper tail. These are in-sample criteria, not held-out prediction results.



Each point is BIC minus the minimum within an outcome, so values near zero are preferred. G_3 has the lowest BIC for both compactness tails and the near-split lower tail. G_4 has the lowest BIC for the near-split upper tail. The three-way G_5 model is not the winner for any outcome.

Results: language-level tail prevalence

Here $\hat{\beta}_R$ is the estimated richness coefficient, $p(\hat{\beta}_R)$ is its nominal coefficient-test p-value, and ΔBIC is current BIC minus the previous model's BIC. TL_1 adjusts for $\log(n_\ell)$; TL_2 additionally adjusts for mean individual term count $\bar{K}_\ell$.

Outcome/tail	Model	Adj. R^2	$\hat{\beta}_R$	$p(\hat{\beta}_R)$	BIC change
Near-split low	TL_1: add richness	0.185	-0.0196	0.000002	-18.97
Near-split low	TL_2: add mean terms	0.273	$+0.0021$	0.758	-8.69
Near-split high	TL_1: add richness	0.243	$+0.0182$	$< 10^{-6}$	-22.68
Near-split high	TL_2: add mean terms	0.246	$+0.0248$	0.00014	$+3.12$
Compactness low	TL_1: add richness	0.066	-0.0117	0.0078	-2.64
Compactness low	TL_2: add mean terms	0.207	$+0.0166$	0.029	-14.00
Compactness high	TL_1: add richness	0.232	$+0.0229$	$< 10^{-7}$	-25.40
Compactness high	TL_2: add mean terms	0.504	-0.0161	0.0086	-43.48

All TL_2 richness coefficients are conditional on mean individual term count. Negative ΔBIC favours the expanded model; positive ΔBIC favours the previous model.

Interpretation: language-level tail prevalence

- **Near-split, lower tail.** The richness coefficient changes from -0.0196 ($p = 0.000002$) in TL_1 to $+0.0021$ ($p = 0.758$) in TL_2. The initial negative association is no longer evident after adjusting for mean individual term count.
- **Near-split, upper tail.** The richness coefficient remains positive, changing from $+0.0182$ to $+0.0248$. However, adding mean term count raises BIC by 3.12, so that extra term does not improve the penalised comparison.
- **Compactness tails.** The lower-tail coefficient changes from -0.0117 to $+0.0166$, while the upper-tail coefficient reverses from $+0.0229$ to -0.0161 after adding mean individual term count.
- **Interpretation.** The tails are not mirror images, and the richness association depends on whether mean term count is included. Since both vocabulary measures come from the same naming task, these are conditional associations, not causal effects.

Do age and sex work the same way in every language?



Are demographic associations shared across languages?

- **Age differences**
  - Does age have the same association with the speaker's colour-partition measure in every language?
  - Or do age slopes differ across languages?
- **Sex differences**
  - Is the difference between the recorded female and male groups similar across languages?
  - Or does its size and direction vary by language?
- **Why compare the models?**
  - A common slope uses fewer parameters and gives a simpler model.
  - Language-specific slopes add parameters and must improve model fit enough to justify the added complexity.
- **Statistical target**
  - Compare common-effect and language-specific-effect regressions using the same observations.
  - Assess whether demographic associations vary across languages, not whether age or sex causally changes colour perception.

Age slopes: common or language-specific?

Let Y_s be a continuous speaker-level outcome, $A_s = \text{age}_s/10$ age in decades, $S_s = 1$ for recorded male and $S_s = 0$ for recorded female, and $\ell(s)$ the language of speaker s. The β terms are regression coefficients, α_ℓ is a language-specific intercept, and ε_s is the residual.

Baseline model M_{A0}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \varepsilon_s.$$

All languages share the same age slope β_A.

Expanded model M_{A1}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \delta_{\ell(s)} A_s + \varepsilon_s.$$

Adds language-specific deviations δ_ℓ to the age slope. The reference language has $\delta_{\ell_0} = 0$.

With reference coding, $\alpha_{\ell_0} = 0$. The nested F-test assesses whether the added age-slope deviations improve the model jointly.

Sex differences: common or language-specific?

Use the same outcome Y_s, age predictor A_s, language indicator $\ell(s)$ and sex indicator S_s. The coefficient β_S represents the common conditional difference between recorded male and female groups. The terms γ_ℓ allow that difference to vary across languages.

Baseline model M_{S0}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \varepsilon_s.$$

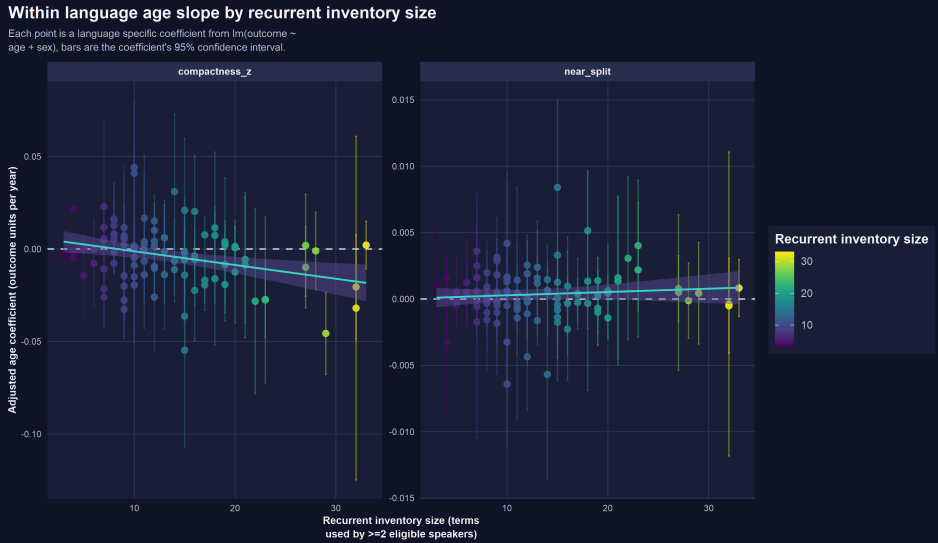
Every language shares the same conditional sex difference β_S.

Expanded model M_{S1}

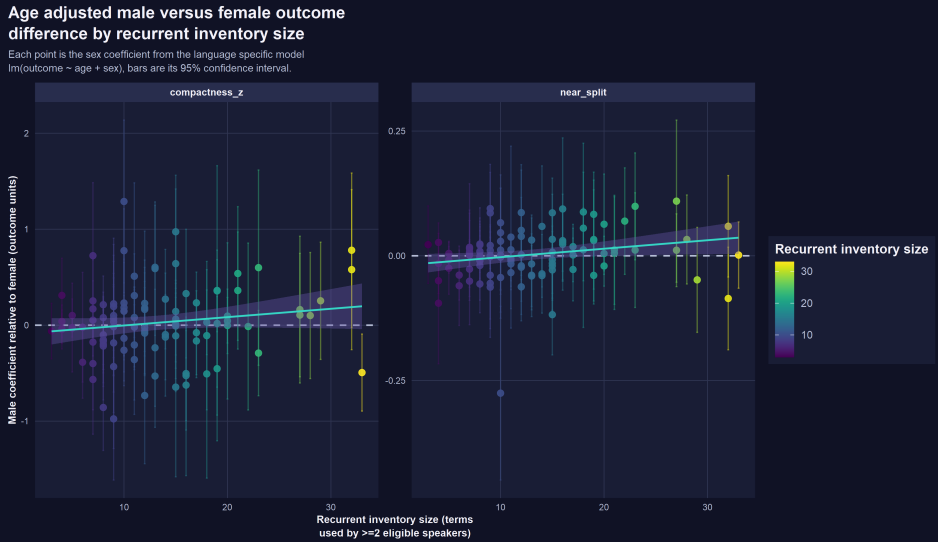
$$Y_s = \beta_0 + \beta_A A_s + \alpha_{\ell(s)} + \beta_S S_s + \gamma_{\ell(s)} S_s + \varepsilon_s.$$

Adds sex-by-language terms γ_ℓ, allowing the conditional sex difference to vary across languages.

With reference coding, $\alpha_{\ell_0} = 0$ and $\gamma_{\ell_0} = 0$. The sex-by-language block is tested jointly. Interpret the F-test alongside AIC, BIC and PRESS to account for the added model complexity.



Each point is a language specific age slope from the within language regression. The horizontal axis is recurrent inventory richness and the vertical axis is the estimated outcome change per decade of age. Error bars show uncertainty, so visual spread alone does not establish heterogeneous slopes.



Each point is a language specific recorded sex contrast plotted against recurrent inventory richness. Error bars show uncertainty. These estimates describe variation across languages and do not by themselves establish that sex differences truly vary by language.



These panels show descriptive speaker level outcome patterns by age and recorded sex before accounting for all language differences. They motivate the common effect and language specific model comparison, but are not themselves a test of heterogeneity.

Heterogeneity results: the test and the penalty tell different stories

Outcome	Extra block	Unadj. R^2 (base $\rightarrow$ expanded)	ΔAIC	ΔBIC	ΔPRESS
Near-split	age slopes by language	.479 $\rightarrow$ .509	+58.7	+627.6	+0.713
Near-split	sex effects by language	.479 $\rightarrow$ .519	+9.3	+578.2	+0.192
Compactness	age slopes by language	.630 $\rightarrow$ .660	+6.5	+575.4	+16.26
Compactness	sex effects by language	.630 $\rightarrow$ .658	+17.0	+585.9	+14.67

These are ordinary R^2 values from the separate common-slope versus language-specific-slope models, which include language-specific intercepts. They are not adjusted R^2 values from the later individual-score E sequence. Positive changes in AIC, BIC or PRESS mean the expanded model is worse on that criterion.

- Some nested F-tests are small. For example, adding language-specific age slopes gives $p \approx 0.020$ for near-split and $p \approx 2 \times 10^{-6}$ for compactness.
- However, these expanded models add roughly 99 parameters, and AIC, BIC and PRESS all deteriorate in the displayed comparisons.
- There is no contradiction: the F-test measures the improvement in in-sample fit, while the criteria penalise complexity or assess leave-one-out prediction.

Does richness matter more for some people than others?



Does the richness association depend on who is speaking?

Age moderation

Is the association between language richness and a speaker's continuous partition measure different for younger and older speakers, after accounting for the age association itself?

Joint moderation

Does the age-by-richness relationship itself vary by sex? This is the role of the three-way age-by-sex-by-richness term.

Sex moderation

Does the richness association differ by recorded sex, rather than merely showing an overall difference between the sex groups?

Decision question

Do these interaction terms improve fit or prediction enough to justify their added complexity, or does a simpler common-effect model suffice?

Does the same moderation appear at the lower and upper tails?

Age-group tail proportions

Across languages, does the older-minus-younger difference in the proportion of low-scoring speakers vary with vocabulary richness? Does the same hold for high-scoring speakers?

Sex-group tail proportions

Does the female-minus-male difference in low- and high-tail prevalence vary with richness across languages?

Two estimands, not one

A regression of language-level subgroup gaps asks whether group differences co-vary with richness across languages. An individual-level interaction asks whether the conditional richness slope changes with age or sex.

A final moderation test

Do the richer-vocabulary associations differ jointly across age and sex, and does the more complex model survive a penalty for additional terms?

Outcome, predictors and interaction ladder

Let Y_s be near-split or compactness, $A_s = \mathrm{age}_s/10$, $S_s = 1$ for recorded male and 0 for recorded female, $R_s = R_{\ell(s),-s}$ be leave-one-speaker-out language richness, and K_s be the speaker's own number of distinct terms.

Exact sequence behind the interaction-penalty results

$$IM_0: \quad Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \beta_R R_s + \beta_K K_s + \varepsilon_s,$$

$$IM_1: \quad IM_0 \text{ plus } \beta_{AR} A_s R_s + \beta_{SR} S_s R_s + \beta_{AS} A_s S_s,$$

$$IM_2: \quad IM_1 \text{ plus } \beta_{ASR} A_s S_s R_s.$$

How the subgroup-gap analysis was constructed

Age-group contrast

Let age_s be recorded age in years. The cut-off $m_{\text{age}} = 34$ is the median age in the analysis sample: $\text{age}_s \leq m_{\text{age}}$ defines the younger group and $\text{age}_s > m_{\text{age}}$ the older group. For language ℓ, let $\bar{Q}_{\ell,O}$ and $\bar{Q}_{\ell,Y}$ be the selected outcome mean (or tail proportion) in the older and younger groups. Define

$$G_{\ell,\text{age}} = \bar{Q}_{\ell,O} - \bar{Q}_{\ell,Y}.$$

Recorded-sex and joint contrasts

Let $\bar{Q}_{\ell,g,h}$ denote the selected outcome mean (or tail proportion) for age group $g \in \{O, Y\}$ and recorded sex $h \in \{M, F\}$. Let $\bar{Q}_{\ell,M}$ and $\bar{Q}_{\ell,F}$ be the corresponding overall sex-specific means/proportions, pooling age groups. Define

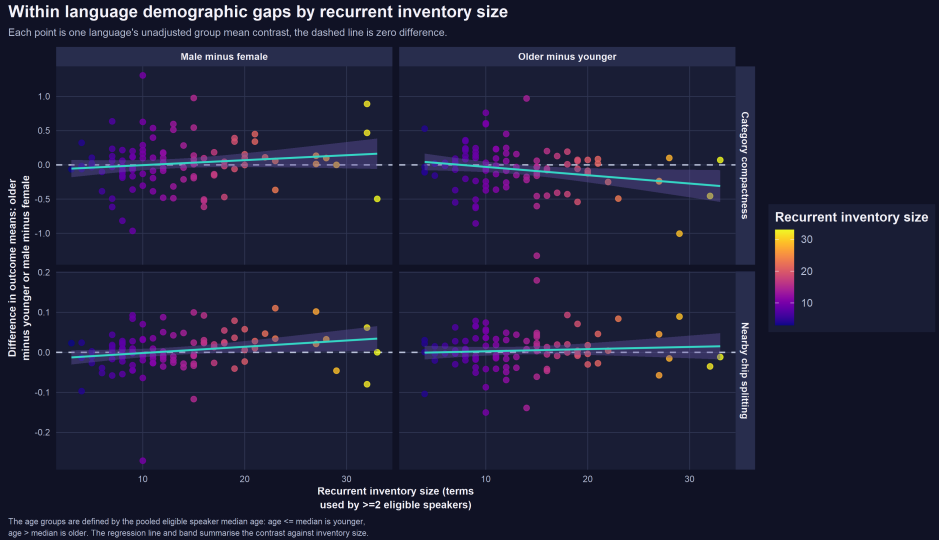
$$G_{\ell,\text{sex}} = \bar{Q}_{\ell,M} - \bar{Q}_{\ell,F},$$

$$G_{\ell,AS} = (\bar{Q}_{\ell,O,M} - \bar{Q}_{\ell,O,F}) - (\bar{Q}_{\ell,Y,M} - \bar{Q}_{\ell,Y,F}).$$

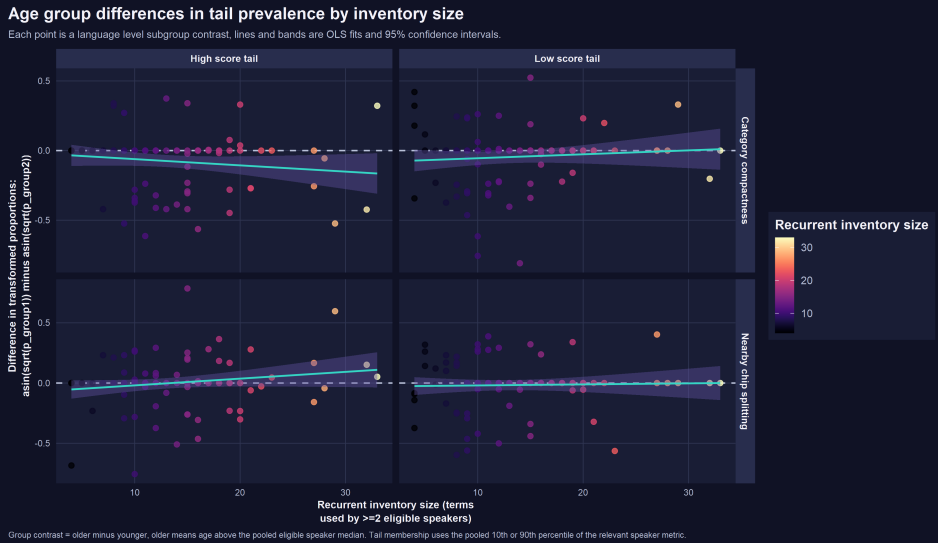
Language-level gap-regression ladder (fit separately for each gap):

- C_0: gap $\sim \log(n_\ell)$ (sample-size baseline).
- C_1: add recurrent inventory richness R_ℓ.
- C_2: add mean individual term count $\bar{K}_\ell$.

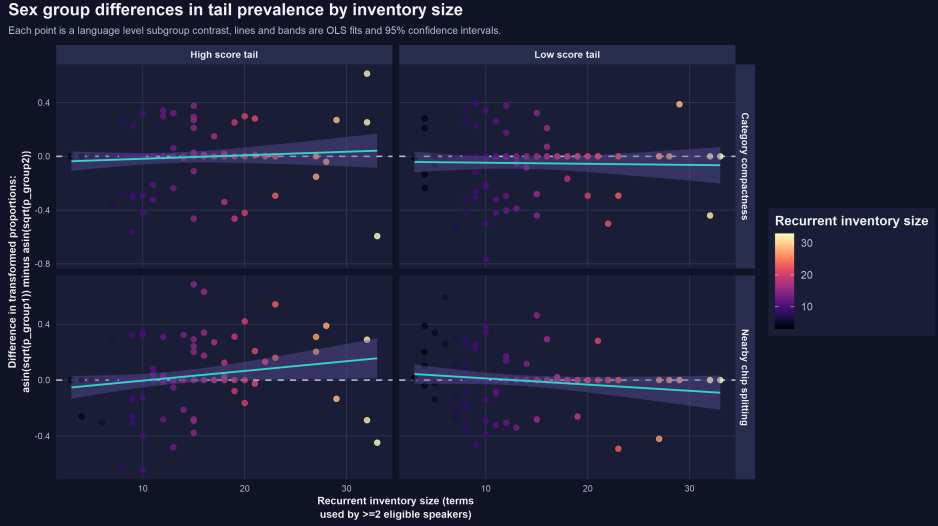
For low/high-tail gaps, the outcome is the difference in the transformed tail proportions. The median split is used only for these language-level subgroup summaries. The earlier age-slope models use age continuously in decades. These aggregate gap regressions are not the same as the speaker-level interaction models above.



Each point is a language level age, sex, or joint subgroup contrast plotted against recurrent inventory richness. The vertical axis is a difference between groups, not an individual score. Group sizes vary, so the pattern is exploratory.



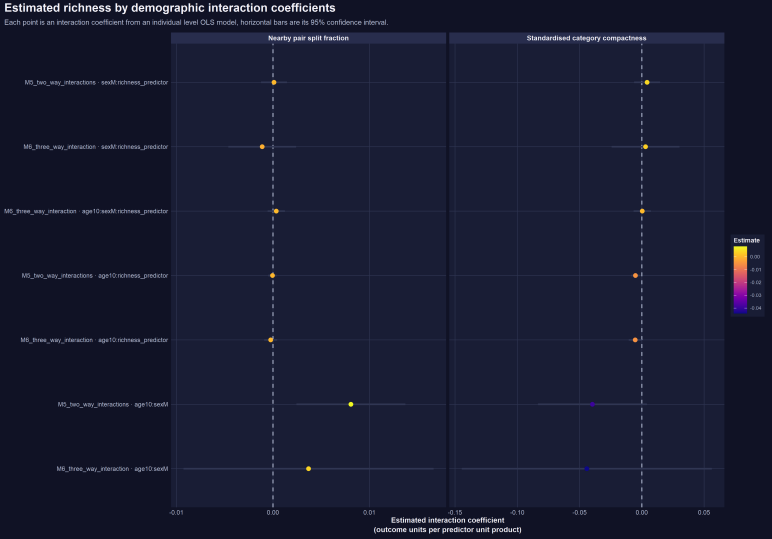
Each point is the language level difference in lower or upper tail prevalence between older and younger speakers. The horizontal axis is recurrent inventory richness. Panels separate the metric and tail so each contrast can be read on its own.



Each point is the language level difference in lower or upper tail prevalence between recorded sex groups. The horizontal axis is recurrent inventory richness. Direction and uncertainty vary across the metric and tail combinations.

What do the subgroup-gap plots show?

- **Continuous outcomes.** The fitted language-level gaps are shallow. For compactness, the male-minus-female line rises slightly with richness while the older-minus-younger line slopes down. For near-split, both fitted lines are close to flat or only slightly positive.
- **Tail outcomes.** Age and sex gap trends vary by metric and tail. Some fitted directions differ between the lower and upper compactness tails, while several near-split gaps stay close to zero. The language-level points are widely scattered and the confidence bands are broad.
- **What we can conclude.** These plots suggest no single demographic gap that changes consistently with richness across all outcomes. They are descriptive regressions of language-level contrasts. The C_0 to C_2 gap-model sequence is defined above, but this deck does not report its AIC, BIC or PRESS comparisons, so we do not declare a formal winner for those models.



The intervals show estimated two way and three way interaction coefficients for age, recorded sex, and leave one speaker out richness. Intervals crossing zero indicate uncertainty about the corresponding interaction. The model comparison determines whether the extra terms justify their complexity.

Moderation results: added interactions rarely pay for themselves

Outcome	Model step	ΔAIC	ΔBIC	ΔPRESS	Reading
Near-split	IM_1 vs IM_0: add two-way interactions	-2.14	$+15.15$	-0.027	BIC rejects
Near-split	IM_2 vs IM_1: add three-way interaction	$+1.45$	$+7.22$	$+0.015$	reject
Compactness	IM_1 vs IM_0: add two-way interactions	-5.72	$+11.57$	-4.18	trade-off
Compactness	IM_2 vs IM_1: add three-way interaction	$+1.99$	$+7.75$	$+1.22$	reject

- BIC penalises each added parameter by $\log n$, about 7.76 when $n = 2355$. The interaction block adds several terms.
- Two-way terms can slightly reduce PRESS or AIC, but BIC rises and three-way terms make prediction worse.
- This table compares the individual-level IM_0 to IM_2 interaction sequence. The subgroup-gap plots above describe language-level contrasts and are not tested by this table.

Where in colour space does richness matter?



Where in the sampled colour space do the patterns occur?

Localising the richness association

Is the relationship between vocabulary richness and the probability of giving different names to nearby chips similar across all sampled colour regions, or is it stronger in some regions than others?

Age and sex by region

Do age-related differences or recorded-sex differences in naming change across regions of the colour chart, even if an overall demographic association is weak?

Interpretation of a regional effect

A regional difference can reveal where the language-level association is concentrated. It does not establish a universal boundary or a perceptual threshold.

How the regions are defined

Regions are operational groups of the sampled chip pairs based on the chart's hue/chroma coordinates. They are not equal-volume regions of all visible colour space.

Region-specific questions and operational regions

Spatial richness. **Does the richness association differ across sampled colour regions?**

Regional demographics. **Do age/sex differences in naming vary by region?**

Let r index a predefined colour region and $\mathcal{E}_r$ be the unique neighbour pairs assigned to region r. Using Y_{si} for speaker s's name for chip i, define

$$D_{sij} = \mathbf{1}\{Y_{si} \neq Y_{sj}\}.$$

Let $\mathcal{E}_{sr} = \{\{i,j\} \in \mathcal{E}_r : i,j \in \mathcal{O}_s\}$ be the pairs in region r for which speaker s named both chips. The regional split rate is

$$Q_{sr} = \frac{1}{|\mathcal{E}_{sr}|} \sum_{\{i,j\} \in \mathcal{E}_{sr}} D_{sij}.$$

Achromatic, achromatic edge, red to orange, yellow to green, green to cyan, blue to violet, purple to red, and cross-band. These group the sampled pairs using mapped hue and chroma coordinates.

Outcome, predictors and model ladder: regional richness

For speaker s and colour region r, Q_{sr} is the fraction of observed neighbouring chip pairs in that region that receive different labels. For language ℓ, the response is the mean regional score

$$\bar{Q}_{\ell r} = \frac{1}{m_{\ell r}} \sum_{s \in \mathcal{S}_{\ell r}} Q_{sr}.$$

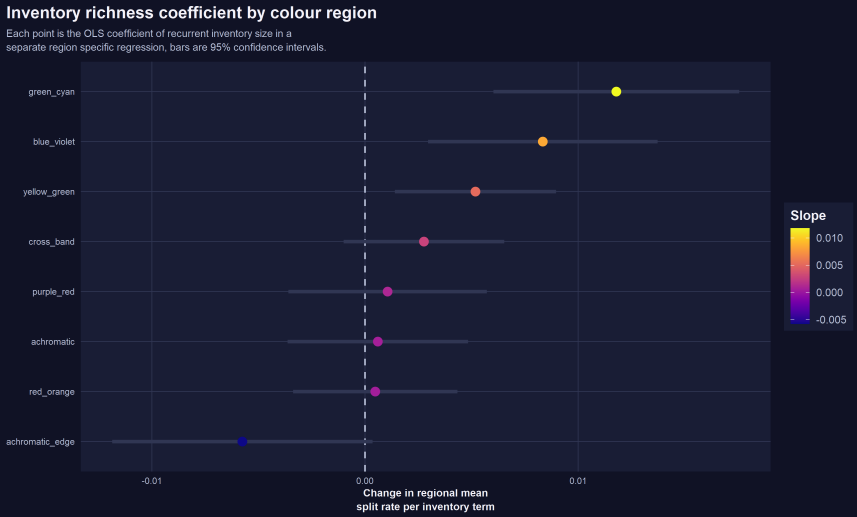
The model is fitted separately for each colour region, with one row per language.

Predictors and added ladder

$$\bar{Q}_{\ell r} = \beta_{0r} + \beta_{nr} \log(n_\ell) + \beta_{Rr} R_\ell + \beta_{Kr} \bar{K}_\ell + \varepsilon_{\ell r}.$$

1. RG_0: intercept only.
2. RG_1: add log eligible speaker count $\log(n_\ell)$.
3. RG_2: add recurrent inventory richness R_ℓ.
4. RG_3: add mean individual term count $\bar{K}_\ell$.

The reported richness coefficient is from RG_3, conditional on sample size and mean term count. The RG labels refer only to this regional OLS sequence.



Each point is the estimated richness coefficient from a separate language level regression for one sampled colour region. Intervals are 95% confidence intervals. Green cyan and blue violet show the clearest positive associations, although eight regional tests make this evidence exploratory.

Results: richness slopes across colour regions

For each region r, $\hat{\beta}_{R,r}$ is the estimated inventory-richness coefficient in the regional OLS model defined on the preceding slide. SE_r is its standard error, and the displayed interval is its 95% confidence interval. The nominal p_r tests $H_0 : \beta_{R,r} = 0$ in that model.

Region r	$\hat{\beta}_{R,r}$	SE_r	95% CI	Nominal p_r
Achromatic	0.0006	0.0021	$[-.0036, .0048]$	.779
Achromatic edge	-0.0058	0.0031	$[-.0119, .0004]$	.065
Red to orange	0.0005	0.0019	$[-.0034, .0043]$	.805
Yellow to green	0.0052	0.0019	$[.0014, .0090]$	.0077
Green to cyan	0.0118	0.0029	$[.0060, .0176]$	.00010
Blue to violet	0.0083	0.0027	$[.0030, .0137]$	.0027
Purple to red	0.0011	0.0023	$[-.0036, .0057]$	.653
Cross-band	0.0028	0.0019	$[-.0010, .0065]$	.149

Row colours identify the colour-space regions. The p_r values are unadjusted for the eight regional tests, so multiple testing should be considered when interpreting significance.

Interpreting the regional richness results

- The estimated conditional richness coefficient is largest in green to cyan (0.0118, 95% CI $[0.0060, 0.0176]$) and next largest in blue to violet (0.0083, 95% CI $[0.0030, 0.0137]$).
- Eight regions were tested. With a simple Bonferroni threshold $0.05/8 = 0.00625$, green to cyan ($p = 0.00010$) and blue to violet ($p = 0.0027$) remain below the threshold. Yellow to green ($p = 0.0077$) does not.
- The regional pattern is suggestive, especially for green to cyan and blue to violet, but should remain exploratory until the regions and contrasts are pre-specified and independently checked.

What the regional demographic plots estimate

For each language ℓ and colour region r, let $\bar{Q}_{\ell r,g}$ be the mean regional split rate among speakers in demographic group g. The plotted quantities are group contrasts, not extra predictors in the regional richness regression:

$$G_{\ell r}^{\text{age}} = \bar{Q}_{\ell r,O} - \bar{Q}_{\ell r,Y}, \qquad G_{\ell r}^{\text{sex}} = \bar{Q}_{\ell r,M} - \bar{Q}_{\ell r,F}.$$

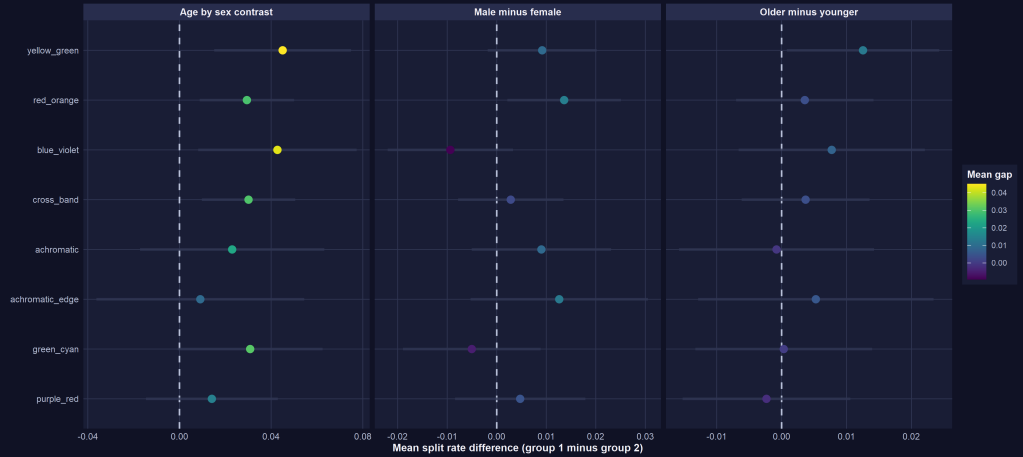
The joint age-by-sex contrast is a difference-in-differences:

$$G_{\ell r}^{\text{age}\times\text{sex}} = (\bar{Q}_{\ell r,O,M} - \bar{Q}_{\ell r,O,F}) - (\bar{Q}_{\ell r,Y,M} - \bar{Q}_{\ell r,Y,F}).$$

- Each point in the contrast plot represents a language-level regional contrast.
- The BIC heatmap compares the RG_0 through RG_3 richness model sequence defined earlier, one region at a time.

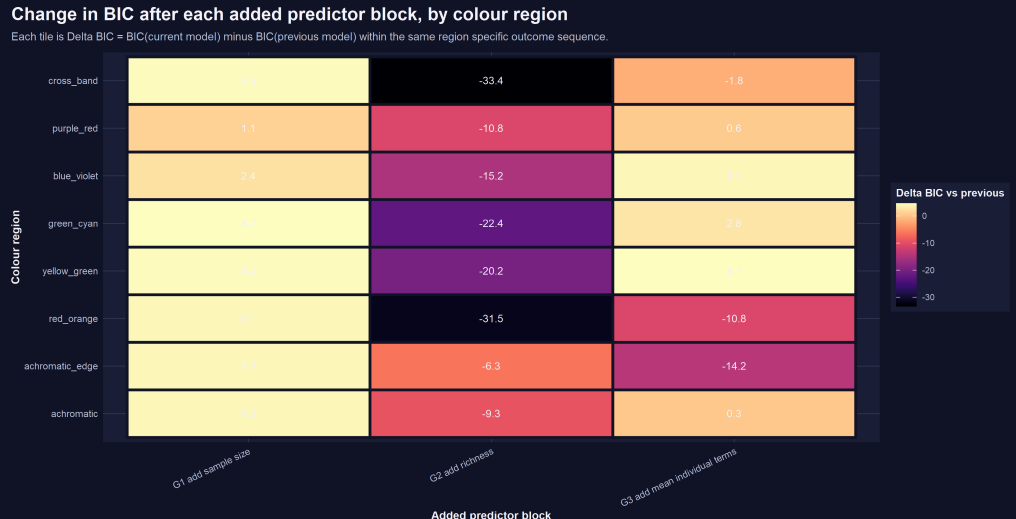
Mean demographic contrast in regional neighbour pair split rate

Each point is the mean language level contrast for a colour region,
horizontal bars show mean plus or minus 1.96 standard errors across languages.



Contrasts are older minus younger, male minus female, or the difference in differences (older maleminus female gap minus younger maleminus female gap). Region specific split rate is the mean fraction of observed k=3 neighbour pairs with different labels.

The points show language level age, recorded sex, and joint subgroup contrasts in regional split rates. Values near zero indicate small average group differences. Scattered estimates and wide intervals signal uncertainty rather than a stable regional demographic effect.



Each tile is the BIC change from the previous regional model, so negative values favour the added block. Adding sample size raises BIC in all regions (+1.1 to +4.7). Adding richness lowers BIC in all regions (−6.3 to −33.4). Mean term count lowers BIC only for achromatic edge, red to orange and cross-band. Its value is region-dependent.

Results: demographic differences across regions

- **Age differences.** The largest older minus younger gap occurs in yellow to green (0.0125, or 1.25 percentage points). Its descriptive interval is $[0.0008, 0.0243]$. The other seven regional intervals cross zero.
- **Sex differences.** The largest positive male minus female gap occurs in red to orange (0.0136, or 1.36 percentage points), with descriptive interval $[0.0021, 0.0251]$. The other seven regional intervals cross zero.
- **Age by sex differences.** The difference in male minus female gaps between older and younger speakers is largest in yellow to green (0.0450), followed by blue to violet (0.0427), cross-band (0.0301) and red to orange (0.0294). The plotted intervals for these four regions lie above zero.
- The age, sex and joint contrasts use 94, 101 and 48 languages respectively. The plotted intervals are means plus or minus 1.96 standard errors across languages. They are descriptive and are not adjusted for multiple testing.

The demographic patterns vary across regions. The clearest descriptive pattern is the positive age by sex contrast in four regions, but these results do not establish a general demographic difference in colour perception.

A better fit is not automatically a better model



Outcome, predictors and model ladders: individual scores

The two continuous responses are fitted separately. Y_s is either the speaker's near-split score or standardised category compactness. A_s is age in decades, S_s is recorded sex, $R_{\ell(s),-s}$ is recurrent richness calculated without speaker s, and K_s is the speaker's own term count.

A. Explanatory model sequence

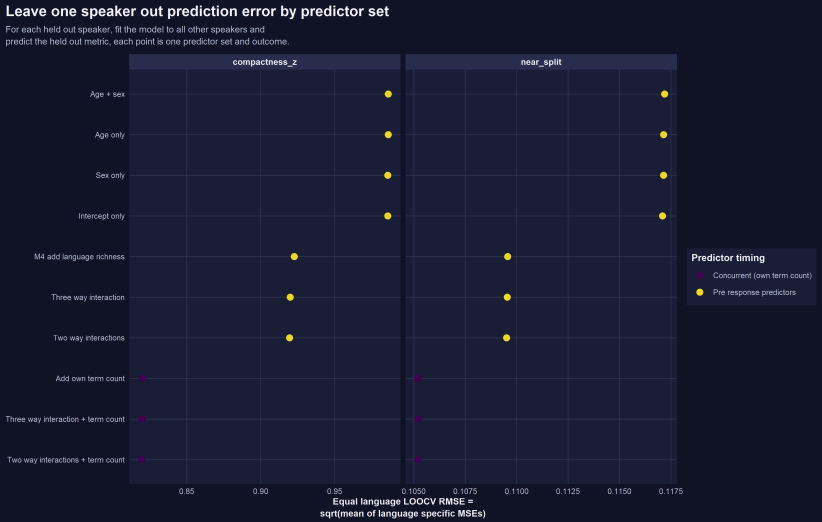
1. E_0: intercept only.
2. E_1: add age A_s.
3. E_2: add recorded sex S_s.
4. E_3: add leave one speaker out richness $R_{\ell(s),-s}$.
5. E_4: add own term count K_s.
6. E_5: add age by richness, sex by richness, and age by sex.
7. E_6: add the three way age by sex by richness interaction.

All steps are shown in the model penalty plots. K_s is concurrent information from the same naming task.

B. Pre response prediction sequence

1. P_0: baseline with age and sex.
2. P_1: add leave one speaker out richness.
3. P_2: add the three two way interactions.
4. P_3: add the three way interaction.

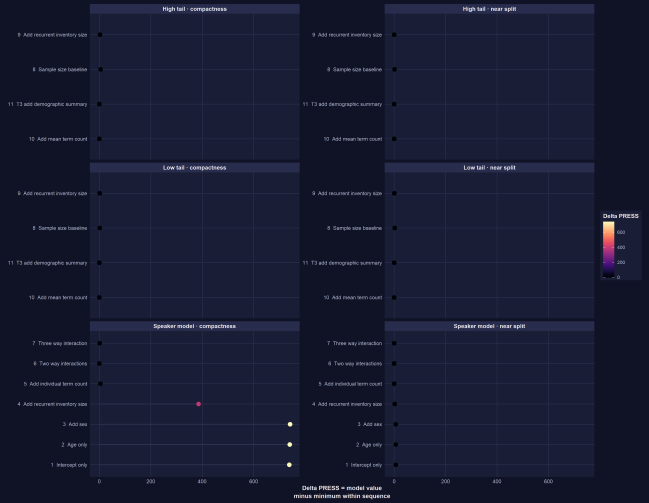
All four steps are compared in the held out speaker results table. Own term count is omitted because it is unavailable before the naming task.



The plot compares held out speaker prediction for near split and compactness. The baseline uses age and recorded sex, the next model adds language richness calculated without the target speaker, and later candidates add interactions. Lower prediction error is better. Richness supplies the main improvement.



Within each outcome sequence, the plotted value is BIC minus the minimum BIC in that sequence. Zero marks the best candidate. Large positive values mean the model is less competitive after accounting for its number of parameters.



The plotted values compare leave one out prediction error within each model sequence. Lower PRESS is better. A candidate can fit the observed sample better while predicting omitted speakers less accurately.

Results: what improves held-out prediction?

Model	near-split RMSE_{LOO}	R^2_{PRESS}	compactness RMSE_{LOO}	R^2_{PRESS}
Age + sex	0.1191	−0.0025	1.0013	−0.0022
Add language richness	0.1115	0.1211	0.9229	0.1486
Add two-way interactions	0.1114	0.1224	0.9200	0.1540
Add three-way interaction	0.1114	0.1219	0.9204	0.1532

- Adding richness reduces RMSE by about 6.4% for near-split and 7.8% for compactness relative to age + sex.
- Two-way interactions improve RMSE by only about 0.08% and 0.32%, respectively, relative to richness alone.
- The three-way model is worse than the two-way model. BIC also rejects the larger interaction block in the explanatory model sequence.

Practical choice: age + sex + leave-one-speaker-out richness as the parsimonious pre-response model.

Defining “best” before choosing a model

- **Explanatory model.** For an observed speaker, the speaker's own term count K_s is available after the naming task. In the individual-score E sequence, the model including K_s has adjusted $R^2 = 19.2\%$ for near-split and 31.4% for compactness. The term count is informative for describing the observed naming partition, but it is derived from that same task.
- **Pre-response prediction.** Before the naming task, K_s is unavailable and must not be used as a predictor. We instead use age, recorded sex and leave-one-speaker-out language richness. The held-out results on the preceding slide show that most of the predictive improvement comes from richness, while interactions add only very small gains.

These adjusted R^2 values belong to the individual-score E sequence. They are not directly comparable with the ordinary R^2 values in the earlier language-specific-slope models, which include language-specific intercepts.

Regression diagnostics: what we checked and why

The following residual, Q-Q and influence plots are from the individual-level OLS fits and the available language-level OLS reference fits. They do not diagnose the ridge/lasso models used for the primary language-mean prediction comparison.

Residuals vs fitted

We looked for curvature, funnels and systematic patterns. Curvature can indicate a mean formula that misses structure, while a funnel can suggest non-constant variance.

Leverage and Cook's distance

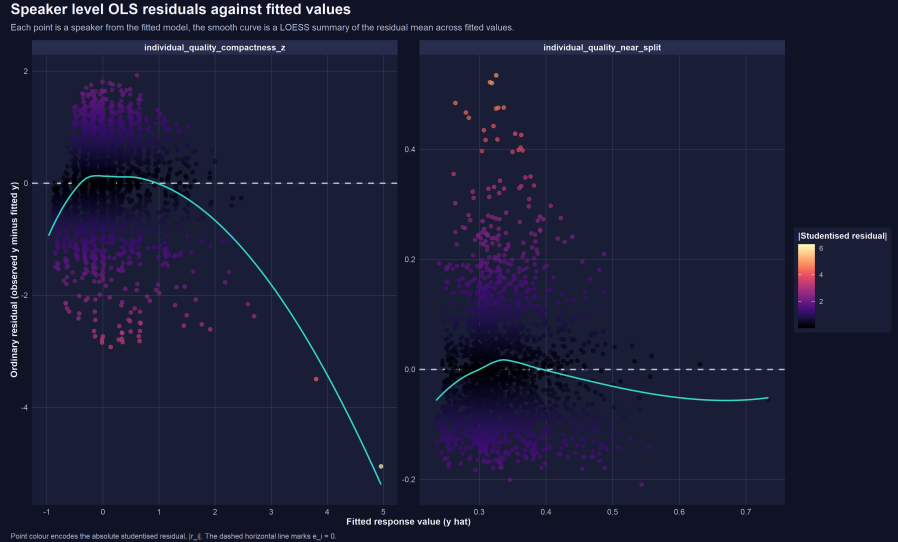
We flagged observations with unusual predictor values, large studentised residuals or high Cook's distance.

Normal Q-Q

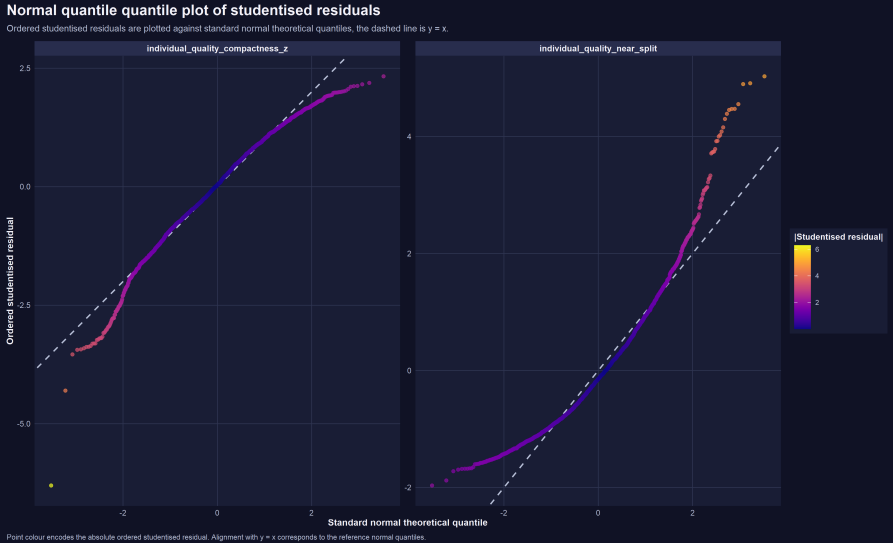
We checked how closely ordered studentised residuals followed the normal reference line. Departures in the tails matter for normal-theory inference, especially in small samples.

Collinearity and VIF

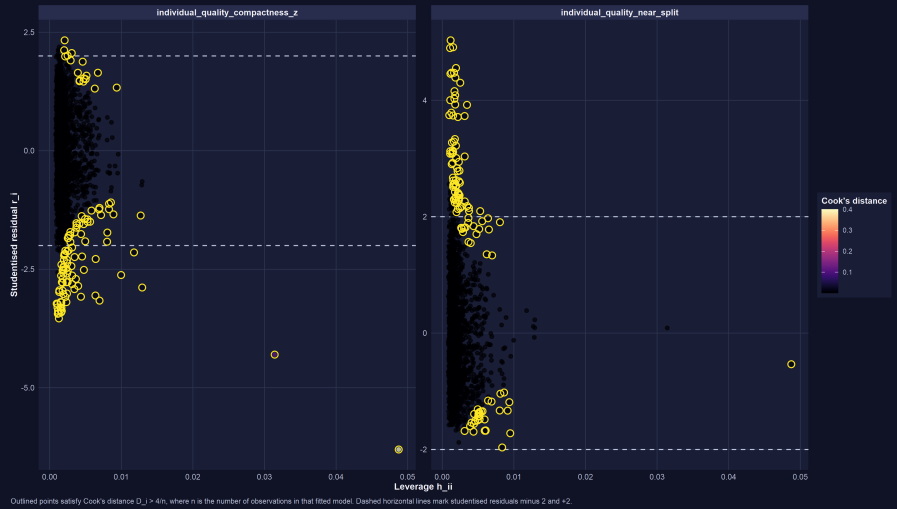
For the simple primary vocabulary sets, VIFs are 3.64 and 3.61 at language level and 3.82 and 3.80 for speaker peer predictors. The interaction-expanded designs are different: VIF reaches 49.03 in the individual near-split model and 9.92 in the language demographic interaction model.



Residuals are plotted against fitted values with a smooth trend. The clearest departure is strong curvature for individual-level compactness, including large negative residuals at high fitted values. Individual near-split also curves mildly. Language-level panels are noisier and show less pronounced structure.



Ordered studentised residuals are compared with normal quantiles. Individual near-split has a pronounced upper-tail departure, while individual compactness has a marked lower-tail outlier. Language-level panels are closer to the line but still show tail departures, so normal-theory inference needs caution.



Outlined points exceed the Cook's-distance screening rule and dashed lines mark studentised residuals of -2 and $+2$. Several individual-level cases are flagged and some language-level points have high leverage. A flag is not a deletion rule. No influence-deletion sensitivity result is reported here.

Main conclusions: what the data say

1. **The penalty.** In repeated nested 5-fold cross-validation, ridge and lasso give the same qualitative result. Adding mean individual term count slightly worsens near-split prediction, but reduces compactness RMSE by about 0.159 to 0.167. Compactness predictive R^2 rises from about 0.19 to 0.52 to 0.53.

2. **Vocabulary predictors overlap, but are not redundant.** Recurrent richness and mean term count correlate 0.831. Their language-level VIFs are 3.64 and 3.61, respectively, indicating substantial overlap but no extreme VIF inflation. Under both ridge and lasso, the compactness coefficient for recurrent richness becomes negative only after conditioning on mean term count.

3. **Tail associations depend on adjustment.** Before adding mean individual term count, richer inventories are associated with fewer low near-split speakers and more high near-split speakers. After adding term count, the lower near-split coefficient becomes close to zero, and both compactness-tail coefficients change sign. The tail pattern is therefore not a single robust effect shared by every specification.

4. **Interactions are expensive.** Some two-way interaction models slightly reduce AIC or PRESS, but BIC often favours the simpler model. Three-way interactions do not improve the model consistently enough to justify their complexity.

5. **Regional patterns are worth investigating.** The largest estimated richness slopes occur in green to cyan (0.0118) and blue to violet (0.0083). These associations remain exploratory because eight regional tests were examined, and the nominal p-values were not adjusted in the original table.

Limitations

Current limitations

- ΔE_{76} is only a proxy for geometric proximity.
- 330 stimuli do not cover the full visible-colour space.
- Age/sex metadata are incomplete and observational.
- Multiple regional tests raise false-positive risk.

References and data provenance

1. Berlin, B. & Kay, P. (1969). *Basic Color Terms: Their Universality and Evolution*. University of California Press.
2. Cook, R., Kay, P. & Regier, T. *World Color Survey Data Archives*. UC Berkeley. `https://linguistics.berkeley.edu/wcs/data.html`
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